closing price on the primary listing exchange was $1,000.01 or more per share, an order for the purchase or sale of an NMS stock of 1 share. The Commission proposed using the IPO price if the prior month's average closing price is not available.

As explained in the Proposing Release, a significant proportion of quotation and trading activity occurs in odd-lots, particularly for frequently traded, high-priced stocks.[240] The proposed definition of round lot would incorporate information about meaningfully sized orders, including many odd-lot quotations in higher-priced stocks that are priced more favorably than the current round lot NBBO, into core data.[241] This would improve the comprehensiveness and

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240 Staff, using the week of June 8-12, 2020, instead of the week of September 10-14, 2018, repeated the analysis from the Proposing Release of odd-lot trade and message volume, duration on the inside, order-book distribution, and quoted spreads for the top 500 securities by dollar volume included in the Proposing Release (see Proposing Release, 85 FR at 16739–40). The results were very similar and confirmed observations discussed in the Proposing Release. Bid-ask spreads widened significantly when calculated using only round lots relative to the odd-lot quotations displayed on proprietary feeds, and as average stock share prices rose, bid-ask spreads based only on round lots generally widened by a greater amount than did spreads based on round lots and odd-lots. Specifically, for the 500 most frequently traded securities by dollar volume, the average bid-ask spread of the 50 securities with the highest share prices decreased (improved or tightened) by $0.19839 when calculated using the proprietary feeds relative to the exclusive SIP feed. Bid-ask spreads for the 50 securities with the lowest share prices showed less improvement when using the proprietary feeds relative to the exclusive SIP feed, decreasing (or tightening) on average by $0.00093. Moreover, frequently traded, high priced securities were more likely to have executions occur in odd-lot sizes (about 34% of the share volume of the 50 securities with the highest share prices) than lower priced securities (about 3.4% of the share volume of the 50 securities with the lowest share prices). Finally, around 91% of the trades that occurred in the two largest securities by market capitalization that have share prices greater than $1,000 occurred in odd-lot share amounts.

241 Staff, using data from May 2020 instead of September 2019, repeated the analysis from the Proposing Release of the proportion of odd-lot trades that occurred at prices that are better than the prevailing NBBO included in the Proposing Release (see Proposing Release, 85 FR at 16740). The results were very similar and confirmed observations discussed in the Proposing Release. During May 2020, a substantial proportion of odd-

usability of core data, facilitate the best execution of customer orders, and reduce information asymmetries.

2. Final Rule and Response to Comments

The Commission received multiple comments on the definition of round lot. Commenters offered suggestions on various ways the proposed round lot tiers should be adjusted and opined on the proposed methodology for determining a stock's price for purposes of assigning a round lot size to a stock.

(a) Round Lot Tiers

Generally: Several commenters expressed general support for a revised definition of round lot for higher-priced securities, but some suggested certain modifications to the proposed definition or otherwise qualified their support.[242] Some commenters agreed with the

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lot trades occurred at prices that were better than the prevailing NBBO. Specifically, approximately 45% of all trades executed on exchange and approximately 10% of all volume executed on exchange in corporate stocks and ETFs (6,926 unique symbols) occurred in odd-lot sizes (i.e., less than 100 shares), and 40% of those odd-lot transactions (representing approximately 35% of all odd-lot volume) occurred at a price better than the NBBO.

242 See, e.g., BlackRock Letter at 3 (stating that the proposed round lot definition is "an elegant solution for increasing odd-lot transparency which innately extends the inclusion of odd-lots to complementary rules and mechanisms such as the determination of the national best bid and offer ('NBBO'), the behavior of order types, and the disclosure of execution statistics" and strikes an appropriate balance between including every odd-lot order and enhancing the quality of market data by establishing a threshold notional amount, but cautioning that round lots should be judiciously calibrated into groups to minimize complexity); Fidelity Letter at 6 (supporting a revised definition of round lot for higher priced securities but urging the Commission to undertake an investor education campaign and to provide sufficient implementation time); Schwab Letter at 4; letter from Joan C. Conley, Senior Vice President and Corporate Secretary, Nasdaq, to Vanessa Countryman, Secretary, Commission, dated July 22, 2020, ("Nasdaq Letter V") at 3 ("Of 31 total comments on the proposed introduction of round lot tiers, fewer than half (14) supported the actual proposal; 11 comments supported a definition different from what

Commission's proposed tiers of round lot sizes,[243] while others agreed with the tier-based approach, but suggested fewer tiers.[244] One commenter, stating that a one-share round lot for stocks over $1,000 could have "unintended negative impacts" on price discovery and routing complexity, suggested that the Commission "defer action" on defining round lots (and including them in core data) until the Commission receives further public input.[245]

Some commenters recommended the Commission eliminate the concepts of round lots and odd-lots in favor of displaying the exact number of shares of every order, arguing that the concepts are "obsolete" because "[m]odern computer processing power is up to the task" of calculating "in real time the cost to buy or sell a given number of shares at the displayed prices."[246]

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the Commission proposed, and 6 opposed it."); letter from Luc Burgun, President and CEO, NovaSparks S.A., to Vanessa Countryman, Secretary, Commission, dated Aug. 7, 2020, ("NovaSparks Letter") at 1; letter from Christopher Solgan, VP, Senior Counsel, MIAX Exchange Group, to Vanessa Countryman, Secretary, Commission, dated Aug. 18, 2020, ("MIAX Letter") at 5–6 (recommending that the Commission periodically review the definition of round lot).

243 See, e.g., IEX Letter at 3–4 (stating that the proposed round lot definition would make round lots "less arbitrary and more comparable across securities … [because] [e]ach round lot tier above the $50 price level would represent a minimum notional value of $1,000, resulting in relatively comparable treatment across securities, regardless of per share price"); letter from Hitesh Mittal, Founder and Chief Executive Officer, BestEx Research, to Vanessa A. Countryman, Secretary, Commission, dated May 21, 2020, ("BestEx Research Letter") at 2; Data Boiler Letter I at 24.

244 See, e.g., T. Rowe Price Letter at 4; Capital Group Letter at 3.

245 See State Street Letter at 3.

246 Angel Letter at 15. See also Nasdaq Letter V at 17–19 (suggesting adding intelligent ticks, in which the standard one cent tick that applies to all NMS stocks would be replaced with tick sizes that vary depending upon the trading characteristics of each such stock, while eliminating round lots).

Several commenters suggested that the proposed five tiers would increase complexity, add confusion, compound costs, create execution-quality challenges, and undermine the usefulness of the proposal, although these commenters offered different suggestions to improve the proposal.[247] One commenter, however, stated that adding tiers would not significantly increase complexity.[248]

Notional Value: Some commenters stated that the definition of core data should include all quotes over a certain notional value.[249] Another commenter recommended basing round lots

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247 See, e.g., Clearpool Letter at 11–12; STANY Letter II at 3 ("A more prudent and nonetheless effective approach to addressing the increased trading in odd-lots, would be to include odd-lot quotations in core data while leaving the definition of round-lot as it currently stands."); Capital Group Letter at 3; Letter from Gerald D. O'Connell, SIG Compliance Coordinator, Susquehanna International Group, to Vanessa Countryman, Secretary, Commission, dated July 27, 2020, ("Susquehanna Letter") at 2 (stating that any benefits of the proposing release "will be impacted by … quoting congestion; … customer confusion; and … gaming strategies."); Nasdaq Letter IV at 10; STANY Letter II at 3–4 (recommending that the Commission conduct a derivative market impact analysis because "STANY is concerned the Commission has not considered the impact and potential for investor confusion when trading options on securities with round-lots quotes in sizes less than the 100-share option contract convention"); TD Ameritrade Letter at 10 ("In the current Proposal for round lot tiers at five different increments, the Firm is also concerned for the potential confusion that may also be posed to investors trading options when contracts remain at 100 shares and the NBBO is quoted in lesser sizes."); letter from Robert W. Holthausen, Professor of Accounting and Finance, and Robert Zarazowski, Managing Director, Wharton Research Data Services, The Wharton School, University of Pennsylvania, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Wharton Letter") at 4 ("A top of book replacement product for TAQ, despite possible cheaper costs from competing consolidators, even if it did not consist of more consolidated market data, would incur significantly greater storage costs than TAQ due to changes in the definition of 'round lot.'") (footnotes removed).

248 See MEMX Letter at 4.

249 See Clearpool Letter at 11–12 (recommending, in the alternative, reducing the number of tiers to three with round lot sizes of 100, 50, and 20); letter from Roman Ginis, Founder, Intelligent Cross, to Vanessa Countryman, Secretary, Commission, dated June 1, 2020, ("IntelligentCross Letter") at 3; ACS Execution Services Letter at 3.

on the notional value of an order, rather than the number of shares, and suggested that the Commission ensure that anything over a minimum threshold qualifies as a round lot, eliminating mixed lots.[250]

Different Threshold: Some commenters suggested increasing the $1,000 price-based threshold but did not suggest a specific number.[251] One commenter suggested, as an alternative to including all quotes above a notional level in core data (as noted above), increasing the price-based threshold to $2,000, stating that "the notional value of the median trade today is about $2,000."[252]

Different Tiers: Multiple commenters offered recommendations to recalibrate the tiers in the proposed round lot definition, ranging from two to four tiers.[253] One commenter, who opposed changing the definition of round lot,[254] suggested that, if the Commission is committed

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250 See letter from Alec Hanson, Founder, AHSAT LLC, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("AHSAT Letter") at 3–5. A mixed lot is an order for a number of shares greater than a round lot that is not a multiple of a round lot (for example, an order for 107 shares). See, e.g., Cboe BZX Rule 11.10.

251 See ICI Letter at 7–8; letter from Stephen John Berger, Managing Director, Global Head of Government and Regulatory Policy, Citadel Securities, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Citadel Letter") at 2; ACS Execution Services Letter at 3; STANY Letter II at 3.

252 See IntelligentCross Letter at 3.

253 See, e.g., TD Ameritrade Letter at 6–7; T. Rowe Price Letter at 4; Capital Group Letter at 3.

254 See TD Ameritrade Letter at 6–7 ("The Proposal for changing the nearly universal 100 share round lot to a price-tiered model appears to be based on some misconceptions. Odd lot trade frequency, which is cited as the justification for the price-tiered model, is not a valid proxy for passive order interest. In reality, trade size is more often dictated by the liquidity-taker than the liquidity-provider and is often a result of algorithmic 'pinging' behavior. TD Ameritrade performed a review in 2019 showing that the increase in odd

to changing the definition of round lot, it should use two tiers: 100 shares for stocks priced under $500 and 50 shares for stocks priced over $500.[255] Another commenter suggested a different definition using two tiers: 100 shares for stocks priced less than $250 and 10 shares for stocks at or greater than $250, where the 10 share round lot would remain unprotected.[256] Another commenter suggested three tiers: 100 shares for stocks priced under $100, 10 shares for stocks priced between $100.01 and $1,000, and 1 share for stocks priced over $1,000.01.[257] Some commenters suggested a different three-tiered definition: 100 shares for stocks priced under $500, 10 shares for stocks priced from $500.01 to $1,000, and 1 share for stocks priced $1,000.01 or more.[258] One of these commenters stated that their "recommended round lot sizes of 1, 10 and 100 shares are ones that are used today and that market participants are accustomed to seeing."[259] Another commenter suggested another three-tiered definition: 100 shares for securities priced up to $50, 20 shares for securities priced between $50.01 and $500, 2 shares for

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lot trades was largely due to small liquidity-taking orders, not small passive orders.") (footnotes removed).

255 See id. at 10–11.

256 See T. Rowe Price Letter at 4.

257 See Capital Group Letter at 3.

258 See Virtu Letter at 3; SIFMA Letter at 9; MFA Letter at 10; Susquehanna Letter at 2–4 (stating that the proposal would be likely to increase significantly, for many high-priced securities, "growth in quote changes, order routes, missed executions, and reroutes from missed executions," but acknowledging that these consequences would be reduced to the extent that fewer tiers are adopted); STANY Letter II at 3 (stating that "[a]mong those members who support a change in the definition of round-lots, there is a decided preference for" this particular three-tiered definition).

259 SIFMA Letter at 9.

securities priced from $500.01 and higher.[260] Another commenter suggested reducing the proposed five tiers to four by collapsing the two-share and one-share highest-priced tiers because there are so few stocks in each of those tiers.[261]

Inadequate Justification for the Proposed Tiers: One commenter stated the Commission failed to explain adequately its rationale for choosing the tiers it chose, suggesting the levels are arbitrary.[262] That commenter also argued that the tiers are "clunky" and could result in large shifts in a round lot size in response to a small change in a stock's price.[263] The commenter further stated that the Commission failed to consider alternatives to the round lot definition, including the commenter's "intelligent tick" proposal, which would vary tick size based upon the trading characteristics of each NMS stock.[264] Additionally, that commenter argued that the tiers would complicate the national market system and "upend longstanding conventions" for market

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260 See Schwab Letter at 4. In a subsequent letter, this commenter provided further support for its suggestion of these three tiers. See letter from Jeffrey T. Brown, Senior Vice President, Legislative and Regulatory Affairs, Charles Schwab & Co., Inc., to Vanessa Countryman, Secretary, Commission, dated Oct. 13, 2020, ("Schwab Letter II") at 1 ("An analysis of orders filled for Schwab clients in the first quarter of 2020 shows that just 5 percent of all odd lot orders are placed for stocks priced greater than $500. Of these orders, roughly the same number of orders are placed for stocks priced from $500 to $1000 and stocks priced greater than $1000. With such a low proportion of orders at prices greater than $500, Schwab believes the additional data provided by the SEC's proposed highest tier would not justify the operational complexity it would create or potential to confuse investors.").

261 See BlackRock Letter at 3.

262 See Nasdaq Letter IV at 14–15. See also Angel Letter at 14 ("The Commission appears to have done little in the way of substantive economic analysis to determine the optimal round lot size as a dollar value. If it had, the proposed dollar value would not be oscillating between $100 and $5,000.").

263 See Nasdaq Letter IV at 14.

264 See id. at 18.

participants and their systems as to how they view and process quotes, especially the convention that one order of a security is generally thought of as 100 shares of that security.[265] Similarly, in a subsequent letter, this commenter stated that the Commission did not undertake "data driven analysis" of the proposed round lot definition, that commenters raised concerns about the complexity of the proposal, and that therefore the Commission cannot determine whether the benefits of the proposal outweigh the costs.[266]

The Commission is adopting a modified definition of round lot (and, as discussed above, is including odd-lots that are priced at or more aggressively than the NBBO in core data).[267] Specifically, the Commission is adopting a four-tiered definition of round lot: 100 shares for stocks priced $250.00 or less per share, 40 shares for stocks priced $250.01 to $1,000.00 per share, 10 shares for stocks priced $1,000.01 to $10,000.00 per share, and 1 share for stocks priced $10,000.01 or more per share. These adjustments are responsive to comments that the proposed five-tiered approach is unnecessarily complex and that the new tiers should be based on a higher notional value threshold. The Commission agrees that the number of round lot tiers

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265 See id. at 17.

266 See Nasdaq Letter V at 4–5.

267 See supra Section II.C.2(b). One commenter suggested an "intelligent tick" regime as an alternative to new round lots. However, the commenter's intelligent tick proposal states that the proposal "attempts to address a discrete set of challenges in the national market system" and that policy makers also need to consider "other, related challenges," including round lots. See Nasdaq, Intelligent Ticks: A Blueprint for a Better Tomorrow (Dec. 2019) at 8, available at https://www.nasdaq.com/docs/2019/12/16/Intelligent-Ticks.pdf. Therefore, the commenter's intelligent tick proposal is not presented as an alternative to adopting new round lot sizes. Changes to the tick size of stocks are outside the scope of the rulemaking and the market data issues the Commission is addressing herein.

should be decreased to reduce cost and complexity, avoid potential confusion among market participants, and promote a smoother transition to the new price-based round lot structure. In addition, the Commission believes that the objective of including additional information regarding orders currently defined as odd-lots in core data to enhance the usefulness of this data to market participants, reduce information asymmetries, and facilitate best execution would still be achieved with the simplified definition that the Commission is adopting.[268]

Moreover, under the modified approach, the new round lot tiers would still be normalized at a particular notional value threshold, albeit higher than the $1,000 notional value reflected in the proposed definition, promoting more consistent treatment of securities of varying prices than the 100-share definition that predominates today irrespective of how much a stock is worth. Commenters submitted data suggesting that average trade and order sizes are significantly higher than $1,000.[269] To confirm those comments, staff evaluated all trades that occurred in 2019 and observed that the average number of shares for all trades was about 193 shares, with an average trade size of $8,842 (excluding auctions, the average number of shares per trade was 178 shares, with an average trade size of $8,068). As a round lot is a trading unit that reflects an order of meaningful size to market participants, and since average trade or order sizes are a reasonable proxy for what market participants consider to be a meaningfully sized order, the Commission

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268 See supra Table 1 (showing that under the four-tiered round lot approach that the Commission is adopting, 0%, 65.35%, 88.28%, and 100% of all corporate stock and ETF volume transacted in a quantity less than 100 shares and at a price better than the prevailing NBBO would be captured in the $0-$250.00, $250.01-$1,000.00, $1,000.01-$10,000.00, and $10,000.01 or more tiers, respectively).

269 See IntelligentCross Letter at 3 ("[T]he notional value of the median trade today is about $2,000"); Virtu Letter at 3–4 (estimating that "average retail trade size between 2007 and the present is around 436 shares or $14,581" but also stating that data from 2019 to present show that the vast majority (over 75%) of all trades are still for less than $10,000); Angel Letter at 17 ("[T]he median trade size is roughly $10,000.").

believes it is appropriate to adjust the notional value threshold of the new tiers upward to $10,000 in response to these comments. The Commission believes that using a round figure that is in line with data provided by commenters and internal staff analysis (i.e., $10,000) will reduce potential investor confusion and implementation cost and complexity.

In addition, as discussed in more detail below, commenters overwhelmingly favored protecting orders in the new, smaller round lot sizes, and the Commission is not adopting its proposal to require protected quotations to be of at least 100 shares.[270] In a market environment where the new round lots are protected, adjusting the notional value threshold upward is appropriate so that order protection under Rule 611, and the applicability of other rules under Regulation NMS, are limited to meaningfully sized orders. Similarly, a higher notional value threshold for the new round lot tiers will prevent orders of a smaller notional value from establishing a new NBBO, which could have added significant cost and complexity to the national market system.

The Commission acknowledges that increasing the minimum stock price for the first sub-100 share round lot tier from $50 to $250 will not improve odd-lot transparency for stocks priced between $50 and $250.[271] However, as discussed above,[272] the Commission is including information about all odd-lots priced at or better than the NBBO in core data, which will counterbalance this loss of odd-lot transparency. In making these choices, the Commission has balanced the competing objectives of: (a) improving the display and accessibility of orders that

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270 See infra Section II.E.2(a).

271 See supra Tables 1 and 2.

272 See supra Section II.C.2(b).